SERVO MOTOR CODE:

```
from machine import PWM, Pin
import time

# Position variables
pos1 = 77  # Starting middle position for mot1 (approx
90 degrees)
pos2 = 77  # Starting middle position for mot2
pos3 = 77  # Starting middle position for motrig
step = 3   # How many duty units to move per click

mot1 = PWM(Pin(32))
mot1.freq(50)
mot2 = PWM(Pin(23))
mot2.freq(50)
motrig = PWM(Pin(15))
motrig.freq(50)

# FIX 1: Lock all motors to starting position
immediately at boot
mot1.duty(77)
mot2.duty(77)
motrig.duty(77)

while True:
    if value == "1":
        if pos1 < 125:
            pos1 += step
            mot1.duty(pos1)
            print("Mot1 Right:", pos1)
        value = ""

    elif value == "2":
```

```
    if pos1 > 25:
        pos1 -= step
        mot1.duty(pos1)
        print("Mot1 Left:", pos1)
    value = ""

elif value == "3":
    if pos2 < 125:
        pos2 += step
        mot2.duty(pos2)
        print("Mot2 Up:", pos2)
    value = ""

elif value == "4":
    if pos2 > 25:
        pos2 -= step
        mot2.duty(pos2)
        print("Mot2 Down:", pos2)
    value = ""

if value == "5":
    if pos3 < 125:
        pos3 += step
        motrig.duty(pos3)
        print("Motrig Right:", pos3)
    value = ""

elif value == "6":
    if pos3 > 25:
        pos3 -= step
        motrig.duty(pos3)
        print("Motrig Left:", pos3)
    value = ""
```

```
    # FIX 2: Slowed down loop to reduce BLE interrupt
chaos
    time.sleep(0.05)
```